Optimal

Welcome

Optimal Clinic, Barrie

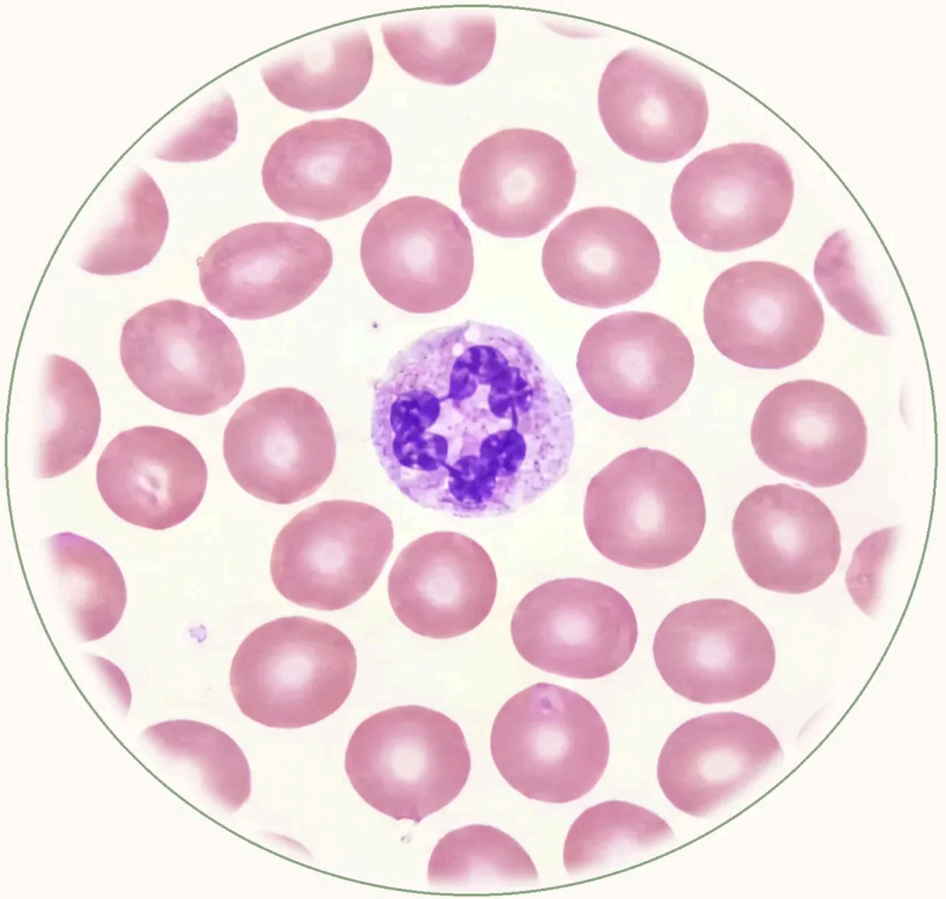
Optimal

EDUCATIONAL TALK · 2026 · BARRIE, ON

A WOMEN'S HEALTH TALK FROM OPTIMAL

Iron deficiency in *women*.

How we measure it, what it does, and how it's treated.



Blood smear, illustrative

FORMAT
Group talk

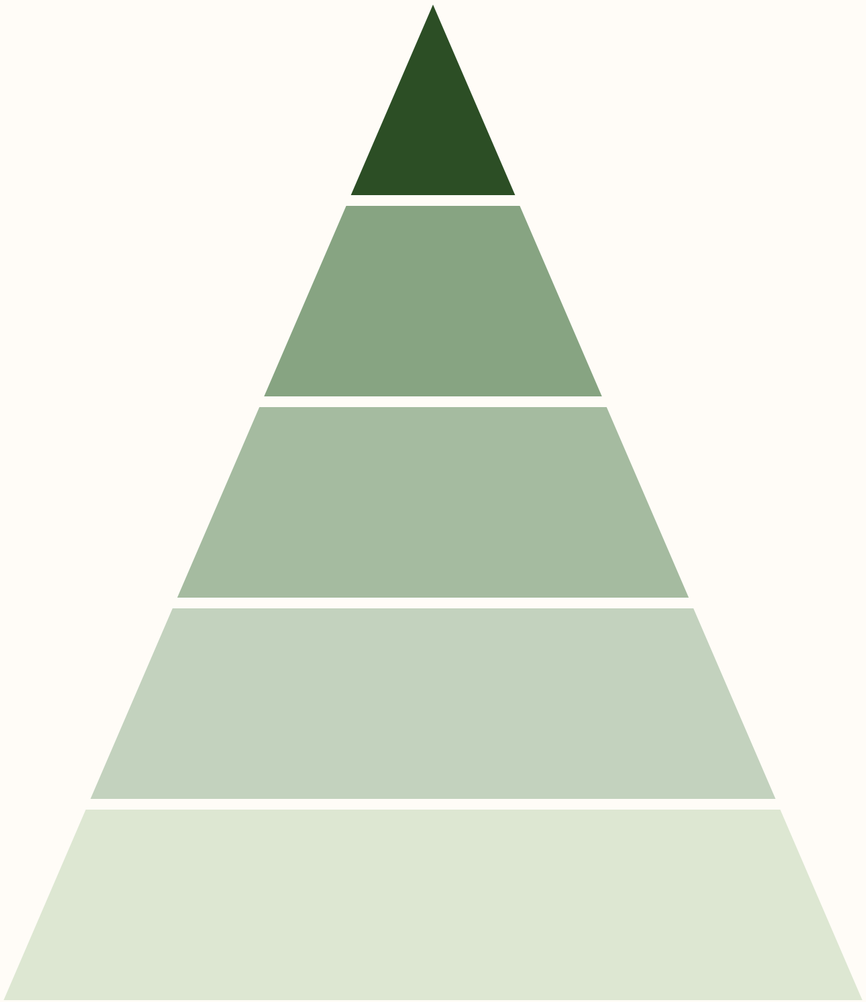
COVERS
Testing to treatment

EVIDENCE
17 sources cited

START AT THE BASE

Supplements sit at the *top*.

Get the base right first. They’re the last layer, not the first.



Nutraceuticals and additional supplements

Protein, carb and fat intake; goal-specific activity

Quantity of activity

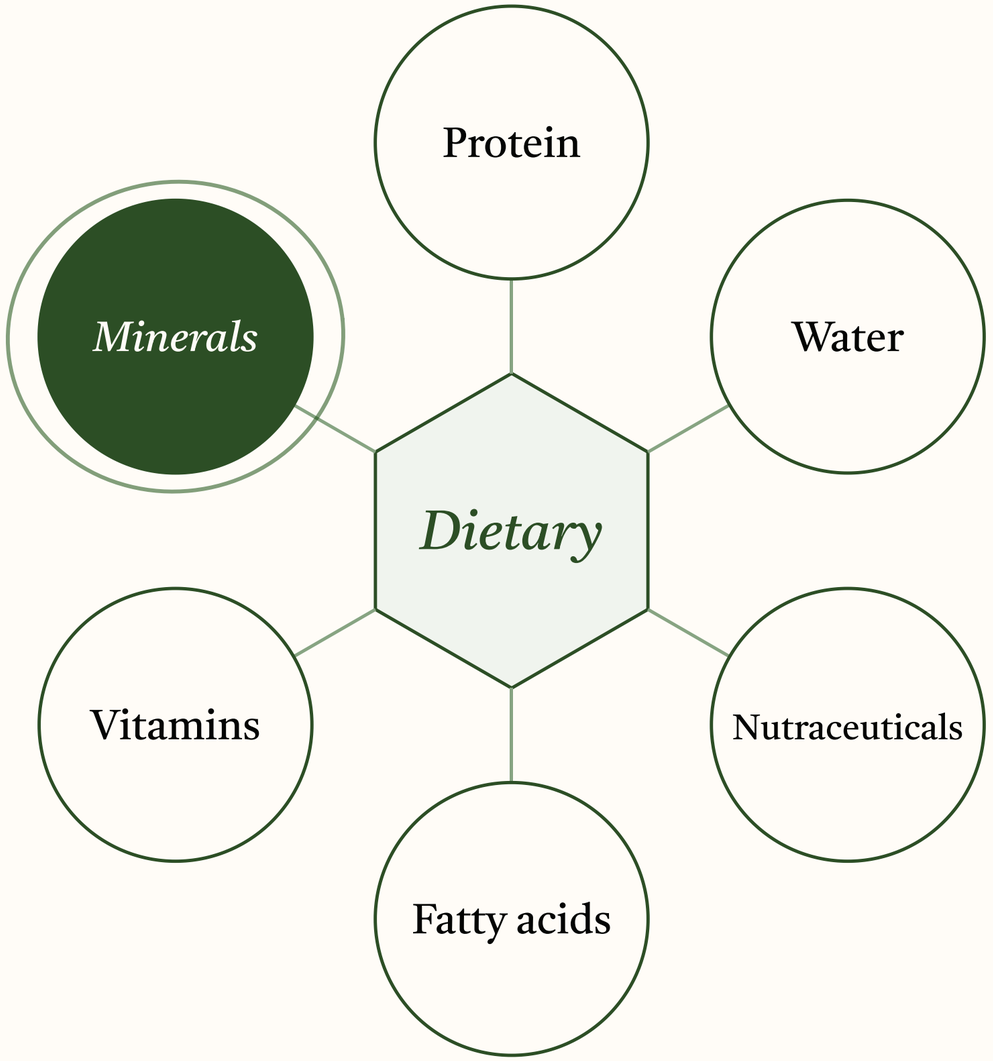
Mental well-being

Adequate rest, caloric intake, hydration and safety

ZOOMING IN

Today, one *mineral.*

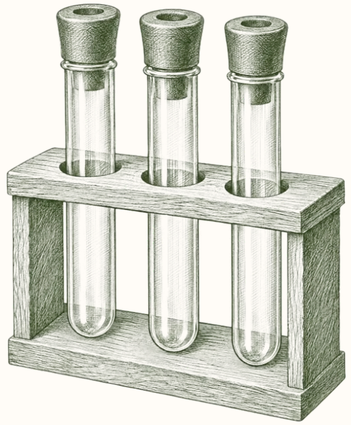
Diet has six building blocks. This talk is about one mineral within them: iron.



TODAY

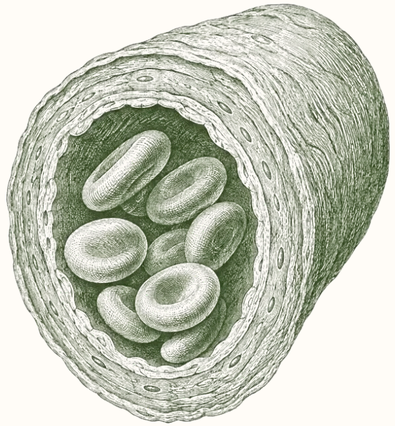
What we'll *cover*.

4 PARTS



01
How we evaluate iron stores

FERRITIN



02
Iron levels

WHAT IRON DOES



03
Consequences of high and low iron

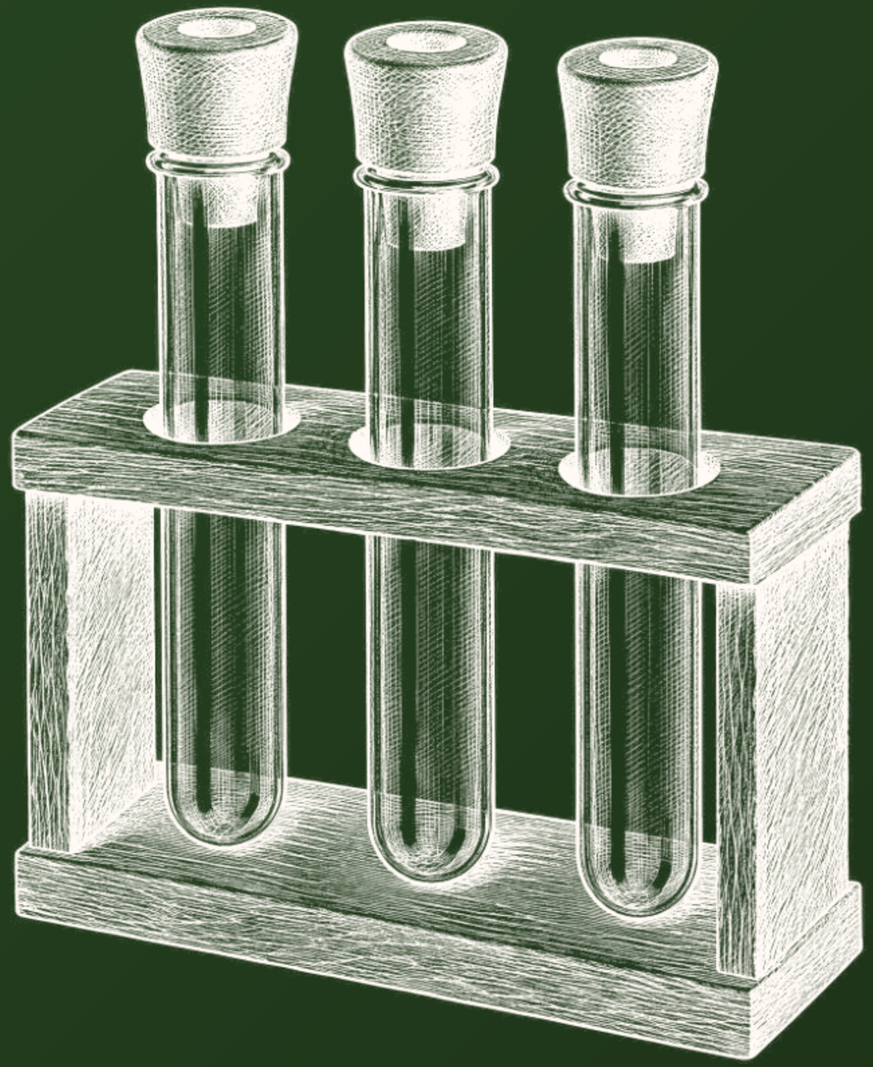
THE EVIDENCE



04
Treatment options

PILL, IV, DIET

How we evaluate iron *stores*.



HOW WE EVALUATE

One test does the *most*.

A blood count picks up anemia. Ferritin tells us about your stores.

Reflects your stores

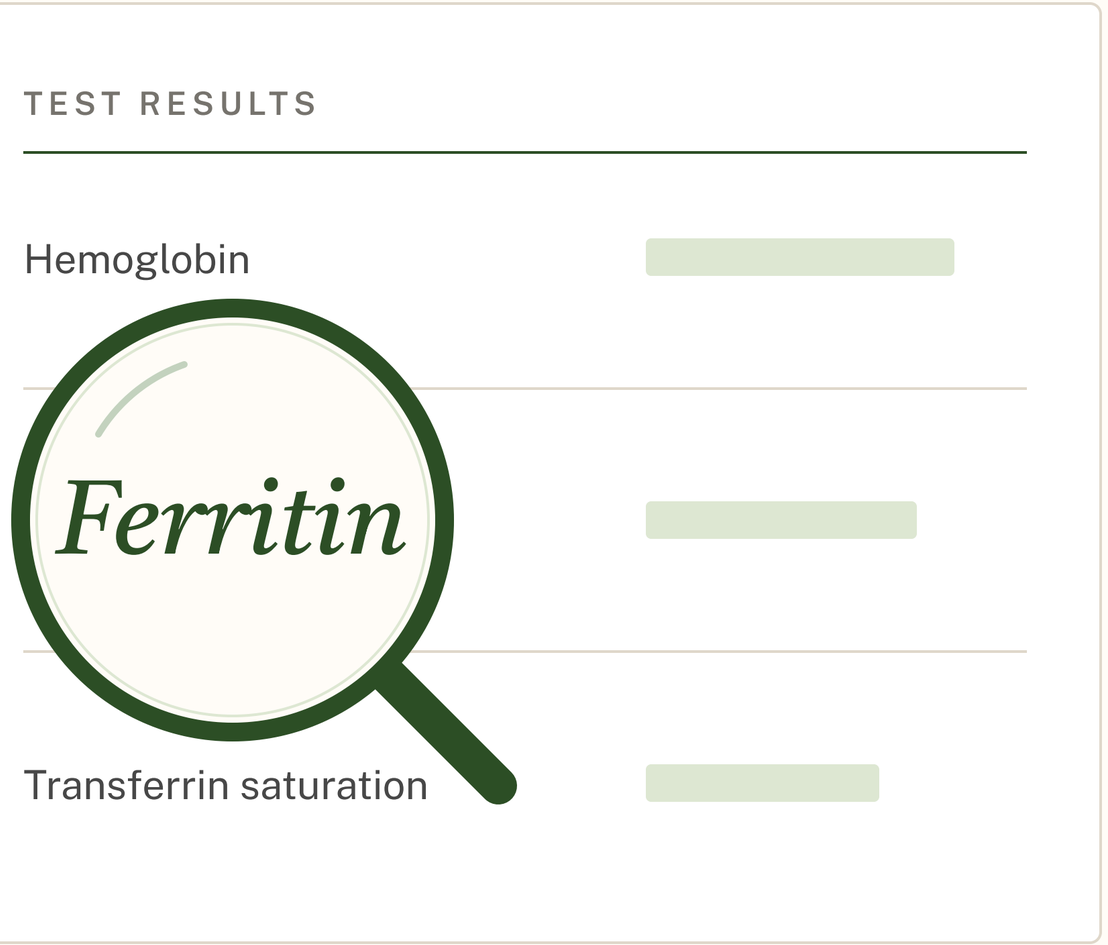
The best screening test for iron deficiency.

Not swayed by your last meal

Recent eating doesn’t change it.

Reads high with inflammation

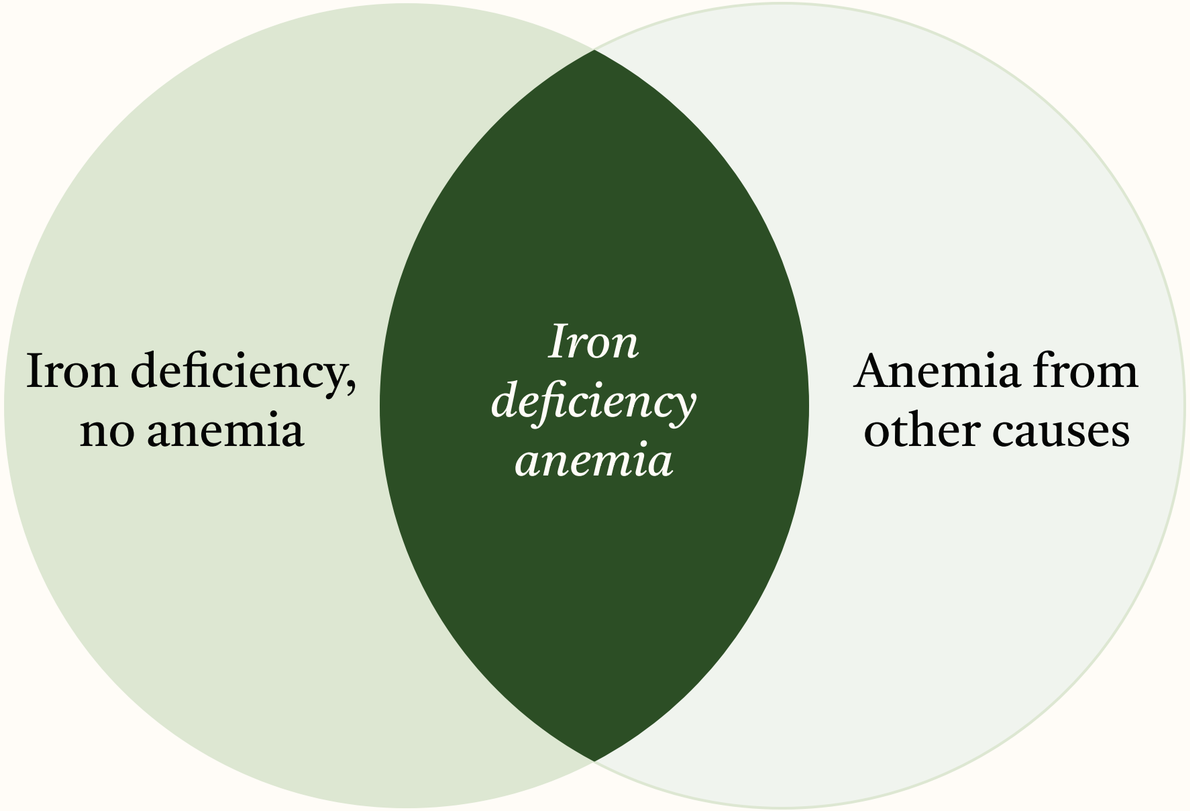
It can look normal even when iron is low.



Source: Sholzberg et al., CMAJ 2025.

HOW WE EVALUATE

Low iron isn't always *anemia*.



38%

of nonpregnant females screened in Ontario had iron deficiency without anemia.

13%

had iron deficiency anemia.

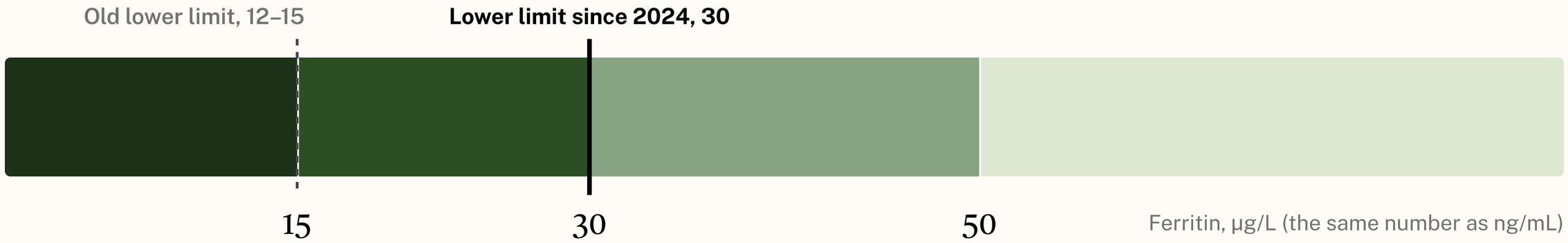
Without anemia, a blood count is typically normal.

Source: Sholzberg et al., CMAJ 2025.

HOW WE EVALUATE

Where the line *moved*.

In 2024, Canada's largest community labs raised the lower limit of normal for ferritin.



Under 15

Flagged low by the old range.

15 to 30

Iron deficiency. Often missed before.

30 to 50

Possible deficiency. With inflammation, check transferrin saturation.

50 and up

The treatment target on oral iron.

TREATMENT IS INDICATED

<50 in pregnancy

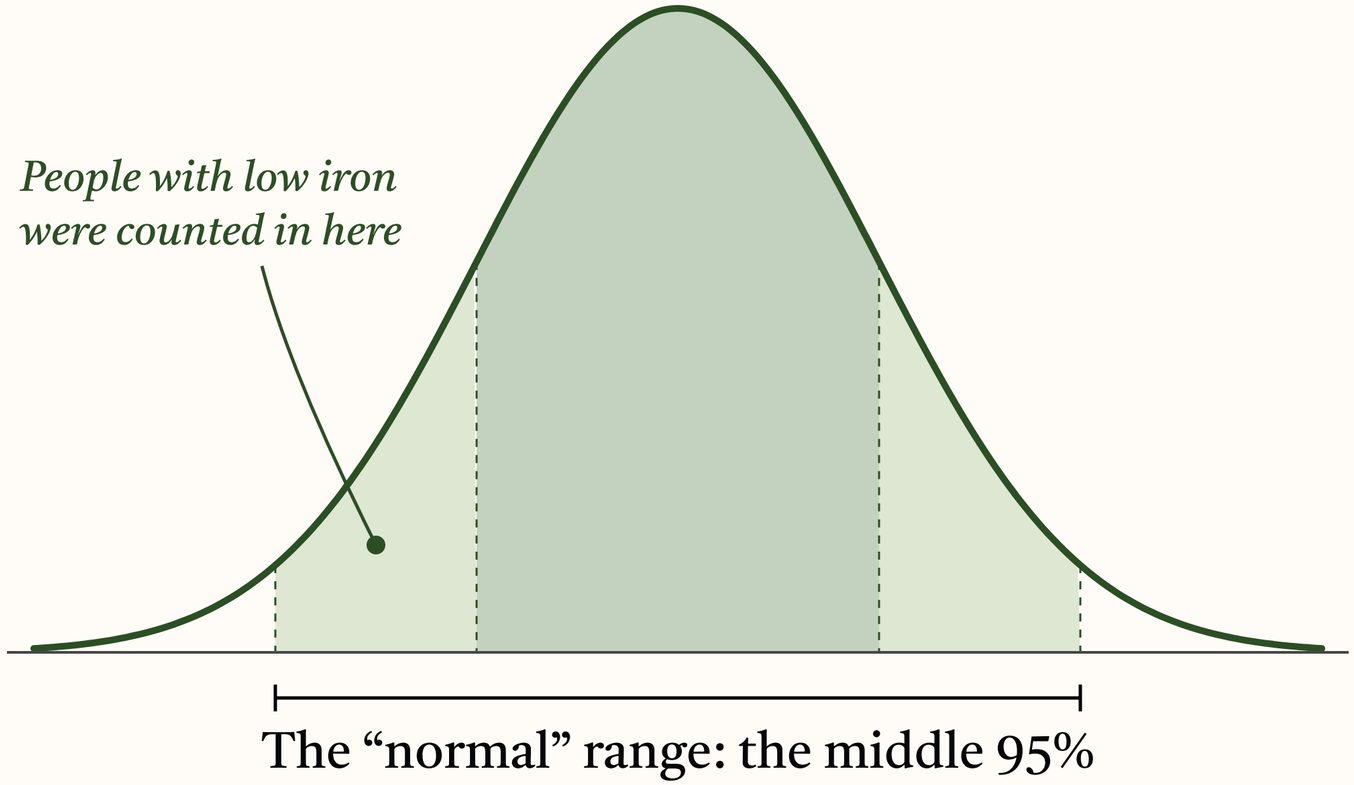
<30 in other adults

<20 in children

Source: Sholzberg et al., CMAJ 2025.

HOW WE EVALUATE

“Normal” is a *statistic*.



It describes the people tested

Not what your body needs.

The old sample wasn’t clean

It didn’t exclude people with iron deficiency.

Twice as common

Iron deficiency without anemia, compared with the 1.2 billion people who have iron deficiency anemia.

Source: Sholzberg et al., CMAJ 2025; Al-Naseem et al., Clin Med 2021. Curve is schematic.

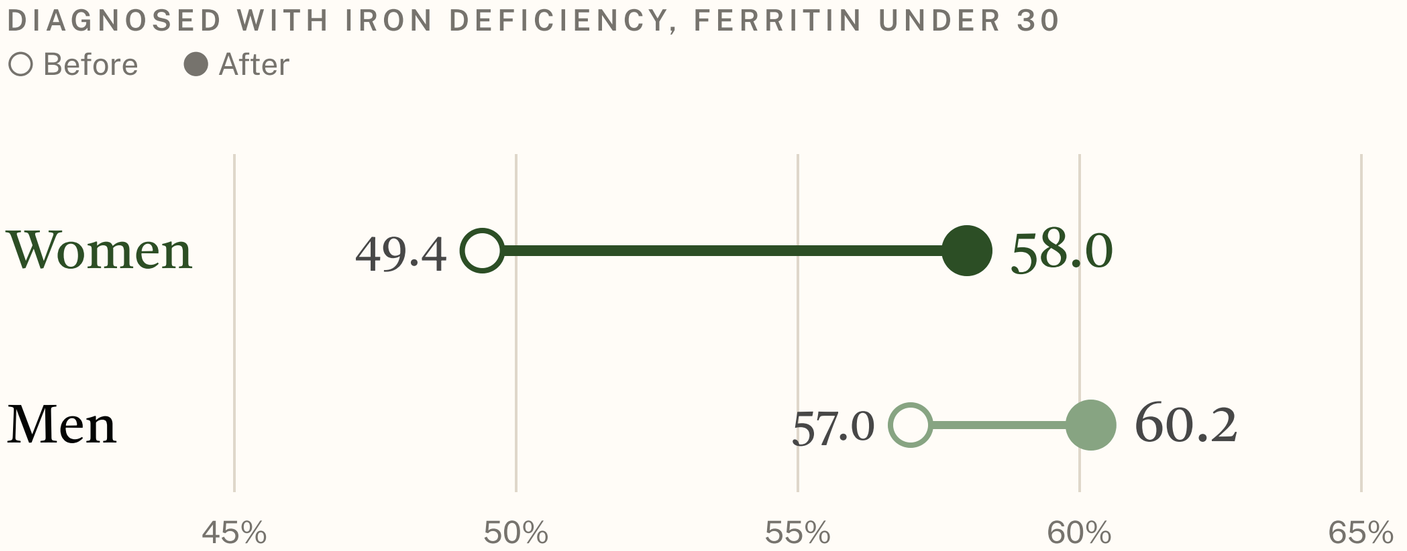
STUDY · 2026

Move the line, find more *women*.

FERRITIN LOWER LIMIT, WOMEN

10 → 30

One US health system changed one number on the lab report. About 50,000 patients later, this is what changed.



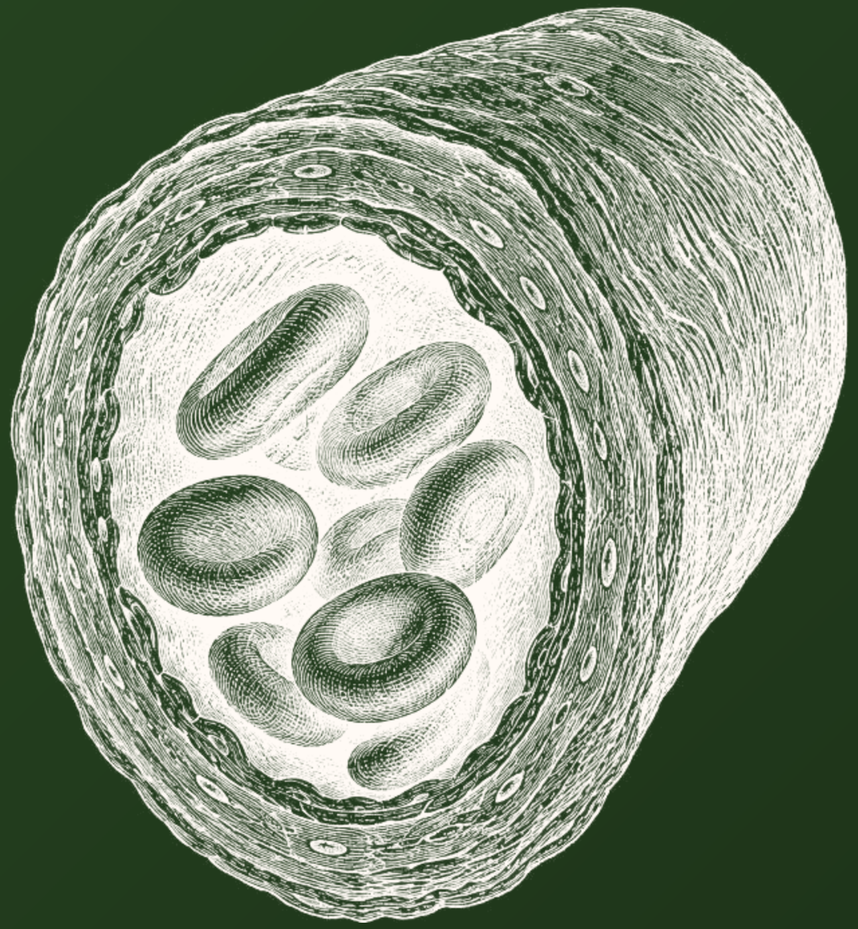
+47%
diagnoses of iron deficiency without anemia

+15%
oral iron prescriptions

+8.2
ng/mL ferritin at 3–6 months in women, vs +4.0 in men

Retrospective pre and post study. Abdelhay A, Reghis M, Salhab T, Jadhav N, Sholzberg M, Kouides P. Impact of optimizing the lower limit of normal for ferritin on iron deficiency diagnosis and treatment patterns: a pre and post-intervention study using EHR data. Am J Hematol 2026;101(3):439–446.

Iron *levels*.



THE JOBS

Iron does the work. Ferritin *holds* it.

- Oxygen transport and storage
- Making blood cells (hematopoiesis)
- Mitochondrial energy production
- Energy metabolism
- DNA synthesis
- Myelination
- Neurotransmitter synthesis
- Cell growth, including bone-building osteoblasts

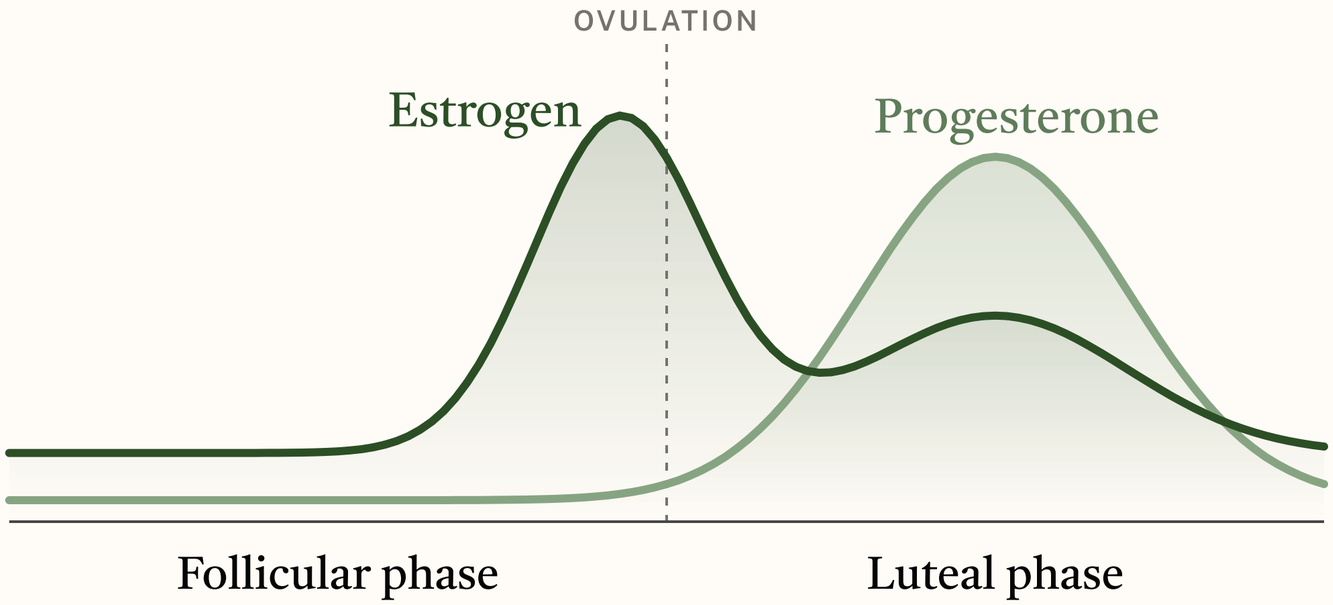
FERRITIN

The store

- The main iron-storage protein
- A buffer for how much iron is available
- Keeps iron usable and non-toxic
- A marker of inflammation, which raises it

STUDY · 2026

Iron moves with your *cycle*.



Estrogen lowers hepcidin, the body's iron gatekeeper. **More iron let in.**

Progesterone raises it. **Less iron let in.**

One result can mislead

Iron markers shift with cycle phase.

A prediction, not yet proof

Sampling across the cycle hasn't confirmed the expected hepcidin peak.

Why perimenopause matters

When these hormones change, iron status can too.

Review. Curves are schematic. Balakirishnan D, Dukovski F, Bytheway H, Dugan C. Cyclical iron homeostasis: the role of sex hormones in female iron regulation. Front Biosci (Landmark Ed) 2026;31(9):54404.

Too much, too *little*.

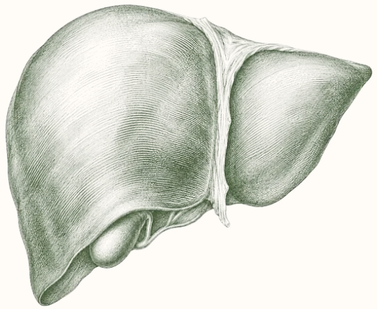


HIGH IRON

More isn't *better*.

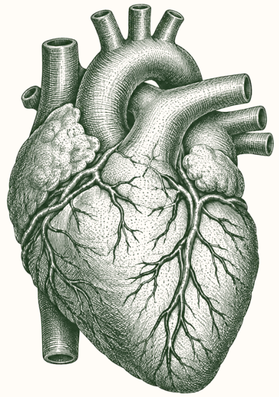
Inflammation can also raise ferritin, so a high result needs a closer look.

CLEAR IN IRON OVERLOAD



Liver

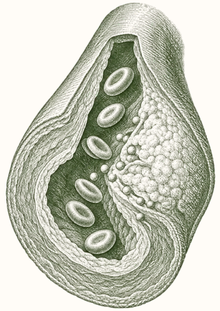
The main store. Overload can scar it over time.



Heart

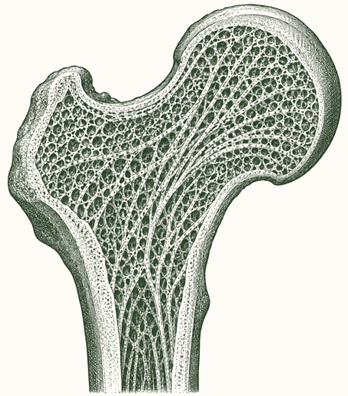
Severe overload can weaken the muscle.

STILL BEING STUDIED



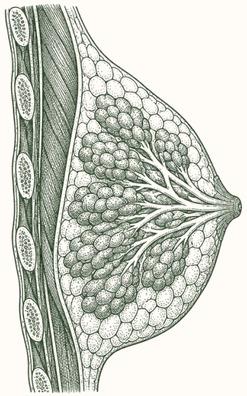
Arteries

The link with cardiovascular disease is still debated.



Bone

Excess iron favours bone loss, especially after menopause.



Breast

A modest link with serum iron. Not shown for ferritin.

Source: EASL guidelines 2022; Ahanchi et al., Clin Nutr 2024; Correnti et al., Biomedicines 2026; Chang et al., BMC Cancer 2019.

LOW IRON

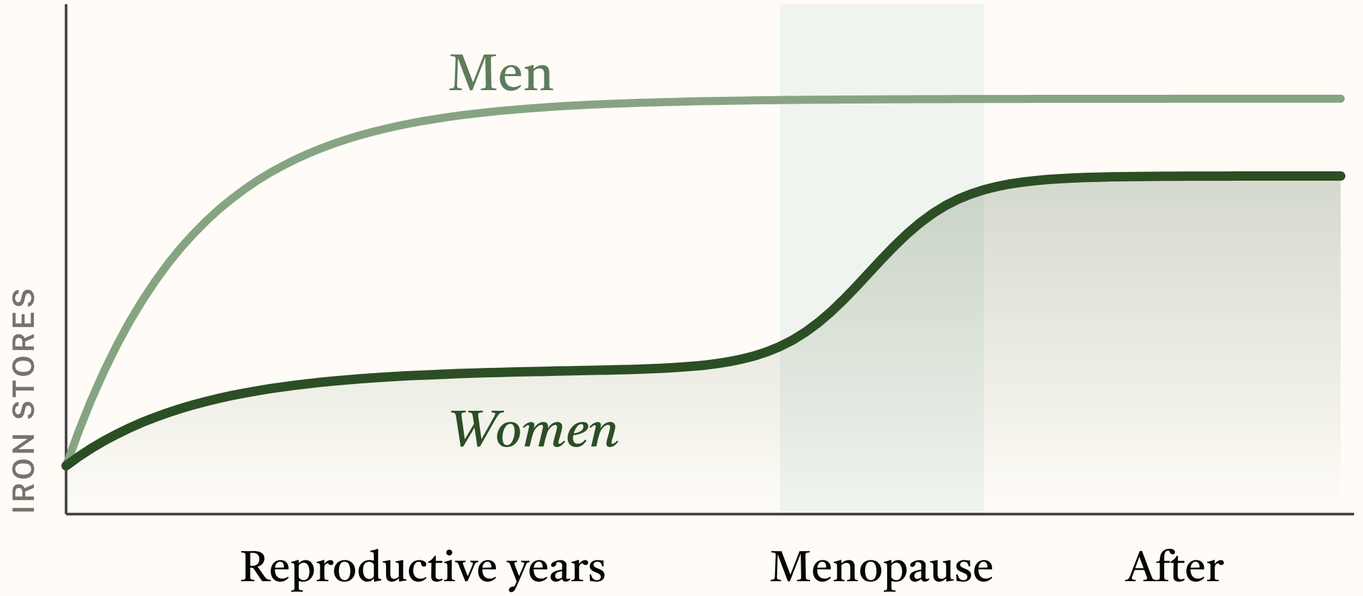
It can show up *before* anemia.

	IRON DEFICIENCY WITHOUT ANEMIA	IRON DEFICIENCY ANEMIA
General	Physical fatigue **83%**, exhaustion, muscle weakness or pain, slower run times, poor sleep	Pallor **45–50%**, fatigue **44%**, muscular weakness
Neurological	Brain fog, low mood or irritability, poor memory and concentration, anxiety, depression, headaches, dizziness	Headaches **63%**, lightheadedness, insomnia, vertigo, fainting **0.3%**
Hair, skin, nails	Hair loss, brittle nails, dry skin	Hair loss **30%**, sore smooth tongue **27%**, spoon-shaped or brittle nails, cold hands and feet
Heart, lungs	Palpitations, chest pain, shortness of breath	Murmur **10%**, rapid heart rate **9%**, palpitations, chest pain, shortness of breath
Also	Cold intolerance **13%**, restless legs **13%**, pica **8%**	Restless legs **24%**

Source: Sholzberg et al., CMAJ 2025. Adapted from Figure 1. Percentages are reported estimated prevalence.

STUDY · 2024

After menopause, stores *climb*.



Lower, then rising

Stores are lower than men’s before menopause, then rise during and after it.

Risk may climb too

Excess iron may add to cardiometabolic risk: more consistently type 2 diabetes than heart disease.

A link, not a cause

Most of this evidence is observational.

Review. Curves are schematic. Ahanchi NS, Khatami F, Llanaj E, et al. The complementary roles of iron and estrogen in menopausal differences in cardiometabolic outcomes. Clin Nutr 2024;43(5):1136–1150.

The evidence is *mixed*.

01 Low iron may matter beyond anemia, and earlier than current cut-offs.

02 Studies are small and cover limited periods of time.

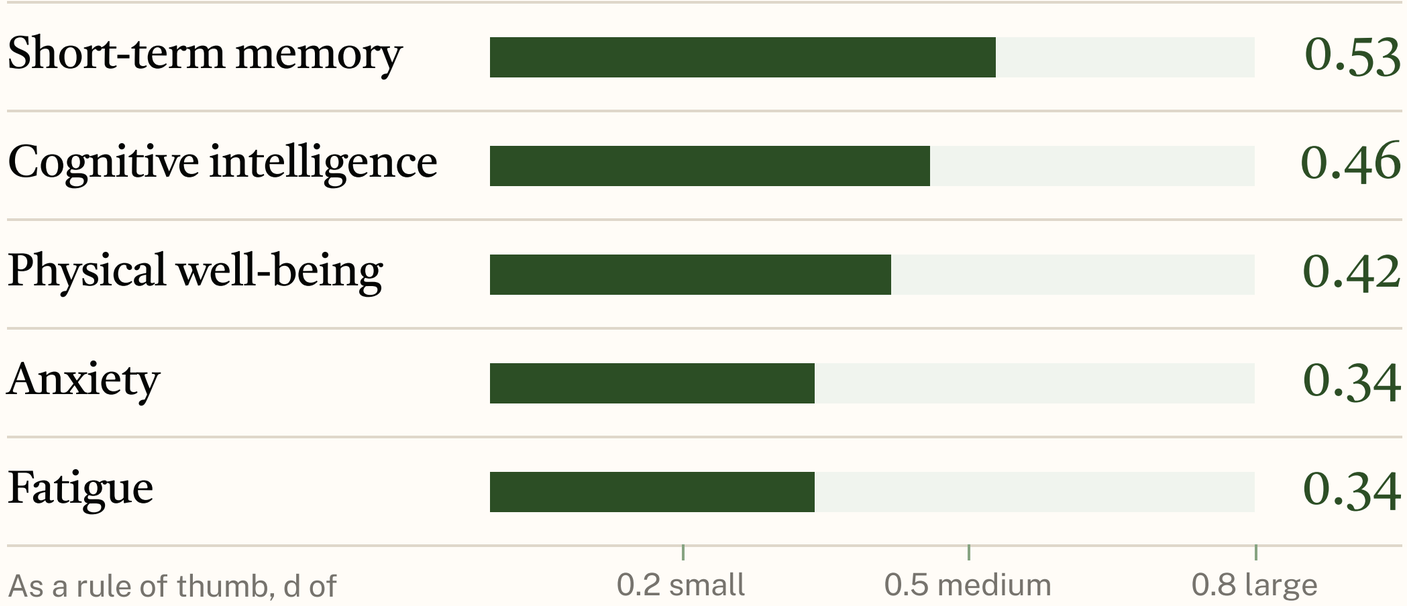
03 Not enough to make strong recommendations for everyone.

04 Enough to explore further, as part of a multimodal approach.

META-ANALYSIS · 2025

Where iron was low, iron *helped*.

RANDOMIZED TRIALS, NO ANEMIA. EFFECT SIZE D



Attention	No improvement
Depression	No improvement in randomized trials

18

studies, 12 of them randomized. 1,408 children, adolescents and menstruating adults.

No effect once iron-deficient participants were excluded.

Meta-analysis. Fiani D, Chahine S, Zaboube M, Solmi M, Powers JM, Calarge C. Psychiatric and cognitive outcomes of iron supplementation in non-anemic children, adolescents, and menstruating adults: a meta-analysis and systematic review. Neurosci Biobehav Rev 2025;178:106372.

STUDY · 2021

Symptoms appear *above* the old cut-off.

FERRITIN
≤50



304 women, 54.9%

Restless legs

2.8 times the odds

FERRITIN
≤20



103 women, 18.6%

Recent hair loss

2.3 times the odds. Also more limits from emotional problems.

FERRITIN
≤15



60 women, 10.8%

Recent hair loss

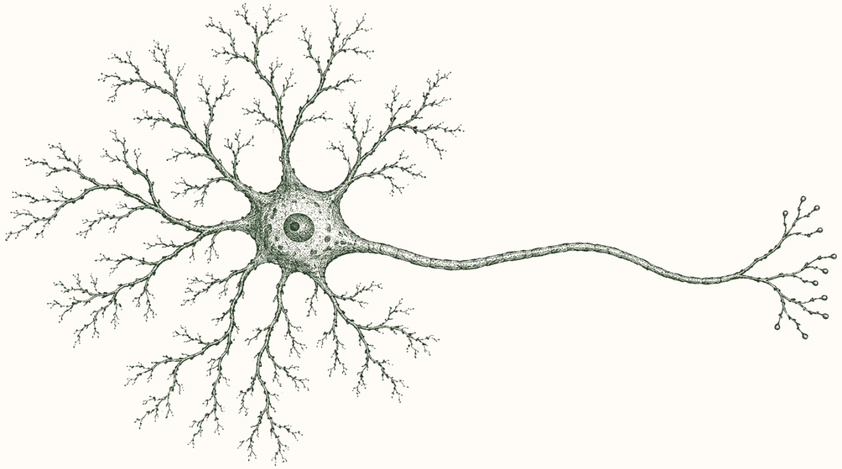
2.2 times the odds

554 non-anemic menstruating women, aged 18–50. A correlation, measured at one point in time.

Cross-sectional study. Beatrix J, Piales C, Berland P, Marchiset E, Gerbaud L, Ruivard M. Non-anemic iron deficiency: correlations between symptoms and iron status parameters. Eur J Clin Nutr 2022;76(6):835–840.

STUDY · 2019

Same score. *Three more minutes.*



+3 min

Median extra time to finish the same intelligence test, with ferritin below 12.

Intelligence and anxiety scores

No difference between the groups.

Activity and endurance

Reported lower with low ferritin.

Brain activity

An EEG pattern linked with withdrawal tendencies.

23 non-anemic women, aged 20–32, in the follicular phase.

Small comparison study. Dziembowska I, Kwapisz J, Izdebski P, Żekanowska E. Mild iron deficiency may affect female endurance and behavior. Physiol Behav 2019;205:44–50.

STUDY · 2016

Better iron, better *focus*. Mostly.

127 non-anemic females, aged 18–35.
A limited study.

BETTER WITH BETTER IRON

Attention

Faster reaction time

Inhibitory control

Steadier reaction time

Planning

Less time needed

NO LINK

Several other executive functions

Possibly a performance ceiling

UNEXPECTEDLY WORSE

Working memory

Appeared to worsen with better iron status

Observational study. Scott SP, Murray-Kolb LE. Iron status is associated with performance on executive functioning tasks in nonanemic young women. J Nutr 2016;146(1):30–37.

REVIEW · 2011

How low does iron have to go before it *matters*?

Who it affects most

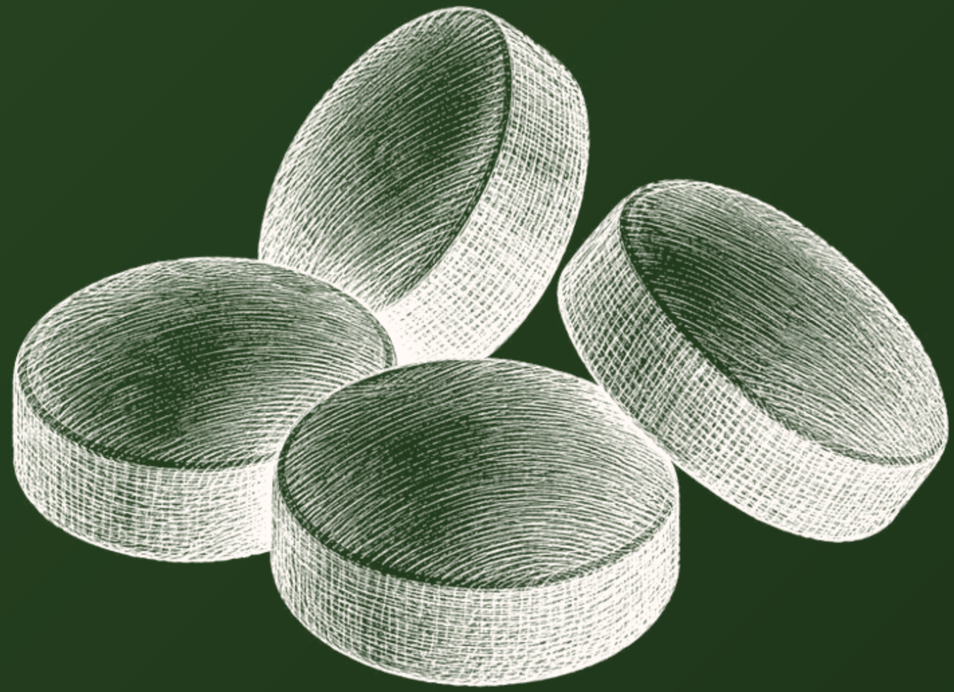
Infants, children and women of reproductive age.

What to keep exploring

How iron interacts with cognition, emotions and behaviour.

Review. Murray-Kolb LE. Iron status and neuropsychological consequences in women of reproductive age: what do we know and where are we headed? J Nutr 2011;141(4):747S–755S.

Treatment *options*.



TREATMENT

Pill *or* IV?

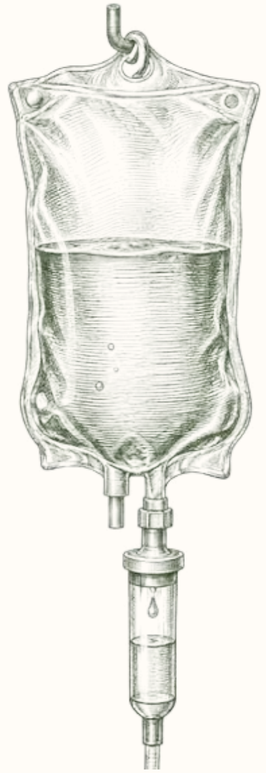


Oral iron

First line, including anemia with hemoglobin above 100

- Hemoglobin improves in days to weeks
- Ferritin improves in about 3–6 months
- Treatment target: ferritin 50 or higher

Gut side effects are what make people stop.



IV iron

When oral iron can’t do the job

- You can’t tolerate oral iron
- A trial of oral iron hasn’t worked
- Malabsorption, or severe anemia

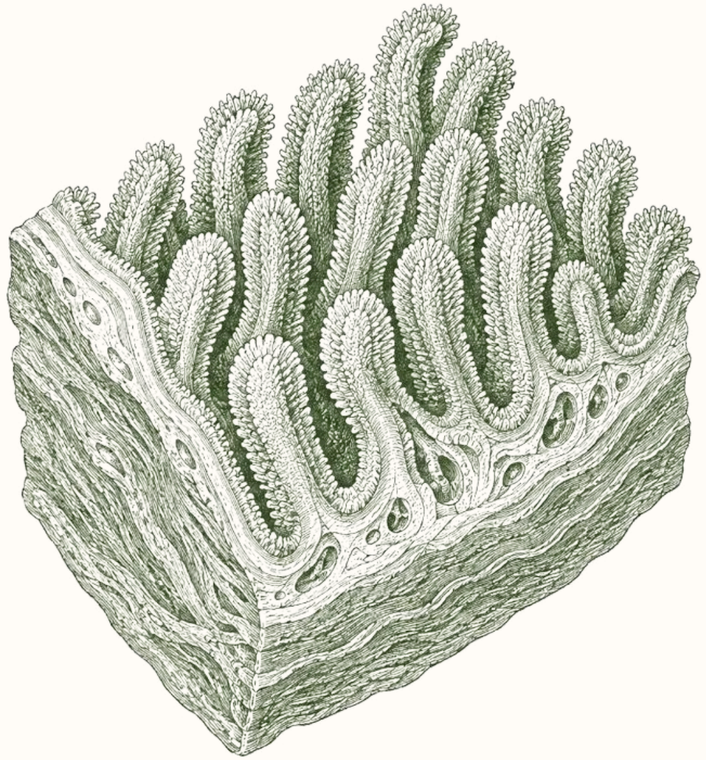
Reactions are uncommon, about 1 in 200, and usually mild. Target: ferritin 100 or higher.

Source: Sholzberg et al., CMAJ 2025.

DIETARY IRON

Food alone won't *fix* it.

An iron-rich diet helps alongside treatment. It won't correct a deficiency by itself.



Heme iron

From animal sources. Better absorbed, and barely affected by what you eat with it.

Non-heme iron

From plants and fortified foods. Absorption depends heavily on what you eat with it.

20 mg of iron a day goes into making red blood cells.

Source: Sholzberg et al., CMAJ 2025; NIH Office of Dietary Supplements.

TREATMENT

Which *iron*?

The best one is the one you tolerate and take consistently.

FORM	ELEMENTAL IRON, PER TABLET	PER DAY	IN SHORT
FIRST LINE: IRON SALTS			
Ferrous gluconate	 35 mg	$0.10	May be better tolerated: less iron per tablet.
Ferrous sulfate	 60 mg	$0.20	May be better tolerated than fumarate.
Ferrous fumarate	 100 mg	$0.25	The most iron of the three, and may be harder on the gut.
COSTLIER, OR MORE SELECTIVE			
Ferrous bisglycinate	 25 mg	$0.30	Fewer gut effects in pregnancy trials. Less data elsewhere.
Polysaccharide complex	 150 mg	$0.75	No robust evidence it works better than salts.
Ferric maltol	 30 mg × 2	—	Prescription, if other oral iron fails or isn't tolerated.

Source: Sholzberg et al., CMAJ 2025; Fischer et al., Nutr Rev 2023; Health Canada approval notice, 2024. Costs are CMAJ's estimates. Ferric maltol isn't in its table.

RECOMMENDATIONS

How to *take* it.

Each tablet contains	
Ferrous sulfate	300 mg
Elemental iron	60 mg

01 Find the elemental iron dose

That’s the number that counts, not the size of the salt.

02 Aim for 60–110 mg elemental iron a day

As your clinician advises. Every other day may be gentler.

03 Take it on an empty stomach

Apart from calcium and magnesium, which interfere with absorption.

04 Take it with vitamin C

Common advice, though one trial found no added benefit.

05 Know the common side effects

Nausea, constipation, diarrhea, indigestion, metallic taste.

Source: Sholzberg et al., CMAJ 2025; Li et al., JAMA Netw Open 2020. Example label for illustration.

TO TAKE HOME

Three things to *remember*.

01 Know your ferritin

Ask for a blood count and ferritin. “Normal” has moved.

02 Look for the cause

Menstrual bleeding is the biggest driver in women. Tell your clinician about heavy periods.

03 Treat low iron, not every tired day

Iron helped where it was low, and too much carries its own risks.

For education only, not medical advice. Please talk with a clinician about your own results.

Questions?

We’re happy to talk anything through, today or after you’ve thought it over.

CALL OR TEXT
(437) 370-0291

EMAIL
care@beoptimal.ca

WEB
beoptimal.ca

REFERENCES

The *sources*.

Scan for the full list, with links.

1 Sholzberg M et al. Diagnosis and management of iron deficiency in females. CMAJ 2025;197(24):E680–E687.

2 Al-Naseem A et al. Iron deficiency without anaemia: a diagnosis that matters. Clin Med (Lond) 2021;21(2):107–113.

3 Abdelhay A et al. Impact of optimizing the lower limit of normal for ferritin on iron deficiency diagnosis and… Am J Hematol 2026;101(3):439–446.

4 Balakirishnan D et al. Cyclical iron homeostasis: the role of sex hormones in female iron regulation. Front Biosci (Landmark Ed) 2026;31(9):54404.

5 Ahanchi NS et al. The complementary roles of iron and estrogen in menopausal differences in cardiometabolic… Clin Nutr 2024;43(5):1136–1150.

6 Fiani D et al. Psychiatric and cognitive outcomes of iron supplementation in non-anemic children,… Neurosci Biobehav Rev 2025;178:106372.

7 Beatrix J et al. Non-anemic iron deficiency: correlations between symptoms and iron status parameters. Eur J Clin Nutr 2022;76(6):835–840.

8 Dziembowska I et al. Mild iron deficiency may affect female endurance and behavior. Physiol Behav 2019;205:44–50.

9 Scott SP et al. Iron status is associated with performance on executive functioning tasks in nonanemic young… J Nutr 2016;146(1):30–37.

10 Murray-Kolb LE. Iron status and neuropsychological consequences in women of reproductive age: what do we know… J Nutr 2011;141(4):747S–755S.

11 European Association for the Study of the Liver. EASL clinical practice guidelines on haemochromatosis. J Hepatol 2022;77(2):479–502.

12 Chang VC et al. Iron intake, body iron status, and risk of breast cancer: a systematic review and meta-analysis. BMC Cancer 2019;19:543.

13 Correnti M et al. The interplay between bone biology and iron metabolism: molecular mechanisms and clinical… Biomedicines 2026;14(2):301.

14 Fischer JAJ et al. The effects of oral ferrous bisglycinate supplementation on hemoglobin and ferritin… Nutr Rev 2023;81(8):904–920.

15 Li N et al. The efficacy and safety of vitamin C for iron supplementation in adult patients with iron… JAMA Netw Open 2020;3(11):e2023644.

16 NIH Office of Dietary Supplements. Iron: fact sheet for health professionals.

17 KYE Pharmaceuticals. Health Canada approval of ferric maltol for iron deficiency anemia in adults (press release). August 27, 2024.